

Electromyographic Evaluation of Automated Penetration Detection in Spinal Drilling

Akari Imai

Institute of Multidisciplinary Sciences
Yokohama National University
Yokohama, Japan
akari.imai@duke.edu

Akito Morishima

Graduate School of Engineering Science
Yokohama National University
Yokohama, Japan
morishima-akito-yp@ynu.jp

Shunya Takano

Institute of Multidisciplinary Sciences
Yokohama National University
Yokohama, Japan
takano-shunya-yf@ynu.ac.jp

Tomoyuki Shimono

Faculty of Engineering and
Institute of Multidisciplinary Sciences
Yokohama National University
Yokohama, Japan
shimono-tomoyuki-hc@ynu.ac.jp

Mitsuru Yagi

Department of Orthopedic Surgery
School of Medicine,
International University of Health and Welfare
Narita, Japan
yagiman@gmail.com

Masaya Nakamura

Department of Orthopedic Surgery,
School of Medicine,
Keio University
Shinjuku, Japan
masa@keio.jp

Abstract—Drilling through the spine during surgery requires precise penetration detection through the bone to prevent injuring the spinal cord. Surgeons typically rely on tactile feedback, which increases physical and mental strain, contributing to musculoskeletal disorders. This study evaluates an orthopedic haptic drill with an automated penetration detection system from the perspective of surgeon-centered benefits. Muscle activity from the posterior deltoid and extensor carpi radialis brevis was examined using surface electromyography during the drilling procedure with and without penetration detection, as well as a commercially available spinal drill. Results showed reduced muscle activation with the penetration detection feature, particularly in the posterior deltoid. This suggests the potential for reductions in surgeon fatigue over time and an increase in their well-being.

Index Terms—haptics, orthopedic drill, spinal surgery, penetration detection, electromyography

I. INTRODUCTION

Cutting and drilling through the spine is a common procedure during spinal surgery. Surgeons must drill through the spine which is composed of hard cortical bone which surrounds the softer cancellous bone [1]. During these tasks, the working surface usually lies above crucial structures such as the dura mater, nervous tissue, and blood vessels; therefore, an accidental breach of the bone can produce severe injuries [2]. A central difficulty during spinal drilling is accurately recognizing the penetration of bone; in practice, surgeons must rely on tactile feedback to judge determine to stop drilling through [3]. This reliance on the their judgment introduces an immense burden for the operator [4], [5].

Spinal procedures can be long and tiring both mentally and physically; surgeons often spend prolonged time doing high-precision and repetitive tasks in static or uncomfortable postures [3]. These physically strenuous tasks that require immense concentration have been associated with musculoskeletal disorders (MSDs) and pain; in fact, 20 – 70% of

surgeons have work-related MSDs [6], [7]. As sustained muscular efforts can lead to increased fatigue, which can impact surgical precision over time, these findings reinforce the need for tools that not only improve technical performance but also lighten the physical load and stress that could accumulate during surgery or throughout the career of a surgeon.

To mitigate the danger of an accidental breach, various studies have sought to automatically detect penetration through bone and stop the drill before excessive puncture. For example, Zakeri and Hodgson analyzed drilling sounds to differentiate between the different bone hardnesses, while Swamy et al. used diffuse reflectance spectroscopy to distinguish between the two bone types [8], [9]. Others enabled surgical robots to predict the bone drilling forces from CT images taken before the operation [10]. Finally, some methods utilized mechanical signals to detect the characteristic pattern change when the bone has been breached. For instance, Osa et al. detected changes in cutting resistance [3].

This study utilized an orthopedic haptic drill previously proposed by Takano et al., which detects penetration by monitoring changes in the thrust force of its linear motor [11]. This haptic drill has shown promise in improving operative safety in spinal surgeries. Despite the extensive emphasis on breach prevention, the benefits for surgeons' well-being are less commonly quantified. Objective physiological measures can enhance this evaluation, such as with electromyography (EMG). EMG records the electrical activity of muscles and has been widely used to quantify fatigue and muscle activation during surgical tasks [6], [12]. This paper utilizes surface electromyography (sEMG), a non-invasive method that employs electrodes placed on the skin's surface [13].

Motivated by the physical burden of surgery and the prevalence of MSDs, this study evaluates the benefit of employing the penetration detection function from the surgeon's perspective using sEMG. Electrodes were placed on the posterior

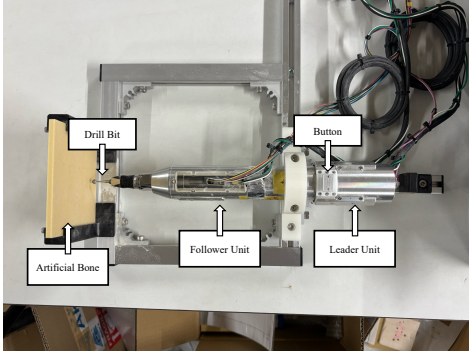


Fig. 1. Structure of the orthopedic haptic drill.

TABLE I
SPECIFICATIONS OF THE MOTOR

Motor Type	Specifications	
	Parameter	Value
Rotary	Size [mm]	$\phi 19.0 \times 115.9$
	Torque constant [mNm/A]	3.67
Voice-Coil	Size [mm]	$\phi 30.0 \times 48.3$
	Force constant [N/A]	19.02
	Continuous force [N]	11.81
	Stroke [mm]	15

deltoid and the extensor carpi radialis brevis (ECRB) to capture differences in muscle activation when drilling with and without the detection feature, as well as with a commercially available spinal drill. The results showed reduced muscle activity, particularly in the posterior deltoid, when penetration detection was enabled, suggesting that this feature may help reduce surgeons' muscle fatigue over time.

The remainder of this manuscript is organized as follows: In Section II, the structure and mechanism of the orthopedic haptic drill are explained, along with a basic overview of sEMG. In Section III, the experimental setup and methods are presented. In Section IV, the results are presented and discussed. Finally, Section V presents the conclusions.

II. THEORY

A. Structure of the Orthopedic Haptic Drill

The orthopedic haptic drill used in this study is shown in Fig. 1. It consisted of two independent units: a leader and a follower. Both units had voice coil linear motors with encoders, and the follower unit also included a brushless DC rotary motor with an encoder. The details of the motor are written in Table I. A drill bit (10BA50DX), coated with diamond particles, was attached to the rotary motor.

The drill was controlled by robust acceleration control, which implemented a disturbance observer and a reaction force observer [11]. Bilateral control was applied to the linear motors in the leader and follower units, and this system is shown in the block diagram in Fig. 2 where the subscripts of l and f denote the leader and follower units, respectively.

Let $\hat{F}$ represent the estimated reaction force. The functions $C_P(s)$ and $C_F(s)$ correspond to the position and force con-

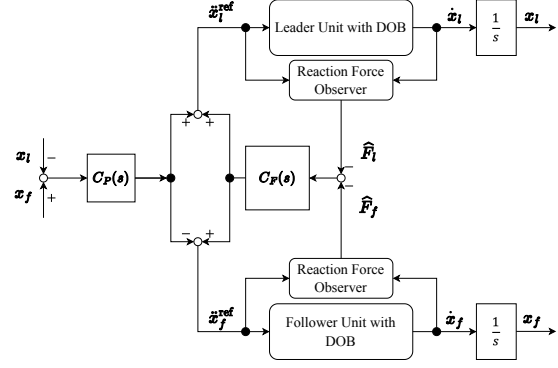


Fig. 2. Block diagram of bilateral control.

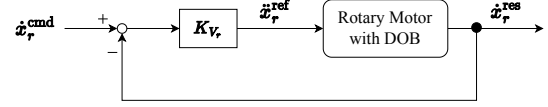


Fig. 3. Block diagram of velocity control for the rotary motor.

trollers, respectively. The mathematical expressions for these controllers are given in (1) and (2).

$$C_P(s) = K_P + K_V \frac{s g_{\text{diff}}}{s + g_{\text{diff}}} \quad (1)$$

$$C_F(s) = K_F \quad (2)$$

In these expressions, K_P , K_V , and K_F denote the proportional position gain, velocity gain, and force gain, respectively. The parameter g_{diff} specifies the cutoff frequency of the pseudo-differentiator.

The rotary motor in the follower unit was regulated through proportional velocity control, as shown in the block diagram in Fig. 3. The subscript r denotes the rotary motor, while K_{V_r} denotes the associated velocity gain. In both block diagrams, the superscripts ref , cmd , and res mean reference, command, and response values, respectively. x , $\dot{x}$, $\ddot{x}$ denote the position, velocity, and acceleration values, respectively.

Bilateral control allowed the leader and follower systems to move simultaneously. When the operator moved the leader unit, they could sense the resistance force from the drill bit due to bilateral control. Furthermore, to rotate the drill bit, the user had to press the button located on the leader unit. This structure and control method also allowed for independence between the two units. Therefore, when bone penetration was detected via the method explained below, the follower unit stopped the breaching of the bone, even if the user continued to apply force on the leader system.

Penetration detection was achieved by monitoring the differential between the follower unit's position and the estimated reaction force. When the drill was moving forwards and the differential reaction force fell below a predefined threshold, the system determined that penetration had occurred. This

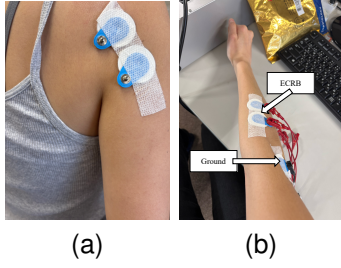


Fig. 4. Placement of the sEMG Electrodes. (a) The posterior deltoid. (b) The ECRB and the ground electrode on the elbow.

threshold was automatically calculated and adjusted based on an estimated viscosity parameter each time. For more detailed information on the drill's control or penetration detection scheme, please refer to a previously published paper [11].

B. Surface Electromyography

Surface electromyography (sEMG) quantifies the electrical activity of skeletal muscles during voluntary contraction [14]. Motor commands from the motor cortex travel along α -motor neurons to the muscle fibers [15]. At the terminal, the neuromuscular junction, a neurotransmitter, acetylcholine, is released, triggering action potentials that propagate along the muscle membrane [15].

A motor unit consists of one α -motor neuron and the muscle fibers it innervates [16]. When a motor neuron fires, it generates a motor unit action potential (MUAP) in its associated fibers. During muscle activation, multiple motor units are recruited; thus, the sEMG signal recorded at the skin surface represents the summed electrical activity of these motor unit action potentials over time.

Raw sEMG amplitude is affected by many factors, such as electrode placement, inter-electrode distance, muscle size, and structure [17], [18]. Thus, direct comparisons between individuals or conditions cannot be made, and raw sEMG signals must be normalized to ensure inter- and intra-subject consistency [19]. The most common method for normalization is by using the maximum voluntary contraction (MVC), measuring the EMG amplitude at a state of maximal effort and then expressing the subsequent EMG signals for each experiment as a percentage of this value (%MVC) [18]. This equation is shown in (3).

$$\%MVC = \frac{RMS_{trial}}{RMS_{MVC}} \times 100\% \quad (3)$$

III. METHODS

A. Muscle Selection

To measure the differences in muscle activity with and without the penetration detection function, the two main upper-limb movements were examined. These were the button releasing movements for rotating the drill bit and the pulling-back movement of the drill handle. Thus, two muscles were selected.

The posterior deltoid was selected for the pulling back movement. It is a superficial muscle located in the back of the

shoulder, making it a suitable muscle for EMG measurement. Because the posterior deltoid is a primary muscle used for shoulder extension, it was predicted to increase at the point of penetration in the absence of breach detection. The electrode placement is shown in Fig. 4(a).

For the button-pressing and releasing movement, the ECRB was selected for analysis. The ECRB is a superficial forearm muscle, allowing relatively easy electrode placement on the skin surface. During drilling with the haptic drill, the ECRB stabilizes the wrist when the operator grips the haptic drill and controls the button. At the moment of bone penetration, the mechanical resistance suddenly decreases, requiring the wrist to maintain stability despite the sudden change in load. Controlled release of the button and stopping the drill bit before excessive penetration also requires precise wrist stabilization. It also plays an important role in wrist stabilization when gripping the drill and controlling the button. Drilling trials using the commercial drill required substantial wrist stabilization, not only during penetration but also throughout the drilling process, to maintain drill stability and accuracy.

Finally, the elbow was selected for the ground electrode location. This was chosen due to its minimal muscle activity [20], [21]. The electrode placements for the ECRB and the elbow are shown in Fig. 4(b).

B. Experimental Setup

sEMG signals were collected using disposable electrodes (Ambu Blue Sensor), which were placed at the target muscle, parallel to the muscle fiber direction. Prior to electrode placement, the skin was cleaned with alcohol to reduce skin impedance. Then, the electrodes were connected to an EMG sensor (P-EMG Plus) with a gain of $1000\times$. The amplified analog signal was then digitized by an A/D board (PCI3178) and transmitted to a computer for data recording. This setup is shown in Fig. 5.

The haptic drill was securely positioned on a surface and placed horizontally. The user was seated with their dominant arm positioned to perform both the pulling back motion and the button-pressing motion. Gloves were also worn to prevent electrical noise from the drill. The setup is shown in Fig. 6. The haptic drill was fixed to an aluminum frame, and it drilled through an artificial bone (Yasojima Proceed, Standard-40). The density of the bone board was $560.8 \pm 56.2 \text{ kg/m}^3$ A

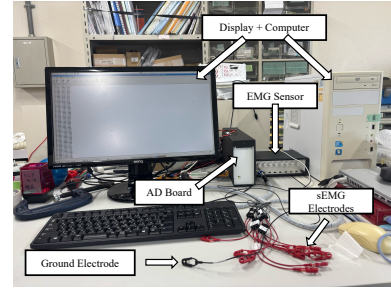


Fig. 5. Experimental system of the computer.

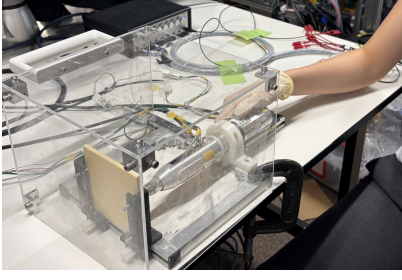


Fig. 6. Experimental setup of the haptic drill.

TABLE II
CONTROL PARAMETERS

Parameter (Unit)	Symbol	Value
Sampling time [s]	Δt	0.001
Drill bit radius [mm]	L_d	2.5
Rotational speed [rpm]	ω	50000
Position gain [s^{-1}]	K_P	6400
Velocity gain: linear [s^{-2}]	K_V	160
Velocity gain: rotational [s^{-2}]	K_{V_r}	160
Force gain	K_F	1
DOB/RFOB cutoff freq. [rad/s]	g_{dis}	150
Pseudo-diff. cutoff freq. [rad/s]	g_{diff}	300
Penetration est. cutoff freq. [rad/s]	g_p	30

diamond drill bit with radius 2.5mm was used, and Table II lists the parameters of the controller. For the third experimental scenario, a commercial handheld drill (Midas Rex MR8) was used, with a different drill bit (MR8-10BA40DX) at a rotational speed of 30000 rpm.

C. Experimental Procedure

To verify that the electrodes were properly attached, the maximum voluntary contraction (MVC) was measured. During MVC measurement, participants were instructed to exert as much force as possible using the target muscles while EMG signals were recorded. The presence of strong and clear signals confirmed that the electrodes were correctly positioned and that muscle activity was being accurately captured.

Each experiment began and ended with measuring participants' MVC to ensure data reliability. Within each trial, participants started and ended in a fully relaxed state to ensure a clear difference between resting and active muscle states.

When penetration detection was enabled, participants drilled into the artificial bone with the haptic drill. Participants maintained a constant posture while pressing the button and advancing the drill forward. The drilling continued until the system autonomously perceived the breach. At that point, participants ceased drilling; timing was disregarded since the drill bit had already been stopped by the system.

In contrast, when penetration detection was disabled, participants relied on haptic feedback and carefully drilled through the artificial bone. When they perceived that the drill had penetrated through, they immediately pulled their arm back and simultaneously released their thumb from the button controlling the drill bit rotation. The procedure was similar

when drilling with the commercial drill, except there was no button on the drill to press.

For the various experimental conditions, with and without penetration detection, eight trials were conducted across one female and three male participants with differing body types.

After collecting the EMG data from the experiments, the raw signals were examined first. As muscle workload is analyzed by looking at the amplitude of EMG, the root mean square (RMS) was calculated to quantify the difference in muscle activation. The normalized RMS values from a 2-second interval around the moment of penetration was observed using (3). Increased normalized EMG was interpreted as greater muscular demand during the task.

In the commercial drill trials, the raw EMG signals for the ECRB showed noise at 0–20Hz with a peak at 2Hz when the drill power was turned off at the moment of penetration. Since a FFT analysis showed that typical EMG frequencies ranged between 40–150Hz, a high-pass filter with cut-off frequency 25Hz was applied to the ECRB EMG signals. In contrast, the posterior deltoid was located further from the drill; thus there was no noticeable noise and high-pass filter was not applied.

IV. RESULTS AND DISCUSSION

A. Raw EMG Analysis

Fig. 7 and Fig. 8 display the raw EMG signals of the posterior deltoid and ECRB, along with the corresponding data of the orthopedic haptic drill, for the cases with and without penetration detection, respectively. The black dotted lines show the moment the system detected penetration. With penetration detection enabled, the drill's movement and the drill bit's rotation stopped synchronously (Fig. 7(c), Fig. 7(d)). However, without penetration detection, the reaction delay of the user can be identified by the gap between the black line and the drill's decrease in position, force, torque, and angular velocity (Fig. 8(c), Fig. 8(d)).

The results showed that the posterior deltoid had low levels of activity when gripping the haptic drill. As shown in Fig. 7(a), when penetration detection was on, there was only noise with the EMG amplitude barely reaching 0.1V, because the muscle was not engaged during gripping tasks. In contrast, the absence of penetration detection required a sudden pull of the drill when the user determined that the bone had been breached. As shown in Fig. 8(a), there was a clearer isolation of muscle activity specifically for the pulling motion that occurred upon penetration. Given the posterior deltoid's primary role in shoulder extension, the elevated EMG amplitude likely reflected this retraction of the upper arm.

The ECRB demonstrated a different activation pattern compared to that of the posterior deltoid. In the two conditions (with and without penetration detection using the haptic drill), raw EMG signals showed initial activation due to wrist stabilization during the sustained button pressing (Fig. 7(b)). The overall EMG amplitude was greater in the absence of penetration detection, suggesting increased muscular effort when the user had to rely solely on haptic feedback. In this case, the button release was associated with a voltage

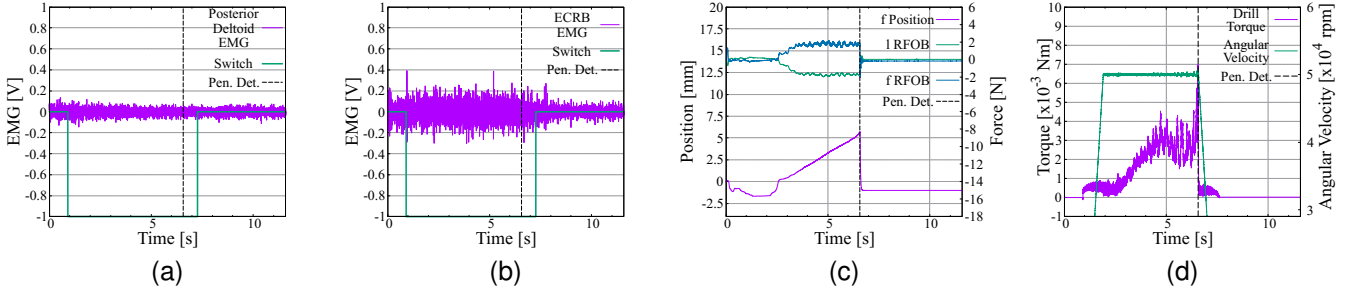


Fig. 7. Results of the drill and the electrodes in the presence of penetration detection. (a) Raw EMG values for the posterior deltoid. (b) Raw EMG values for the ECRB. (c) Position and force of the leader and follower units. (d) The torque and angular velocity of the drill bit.

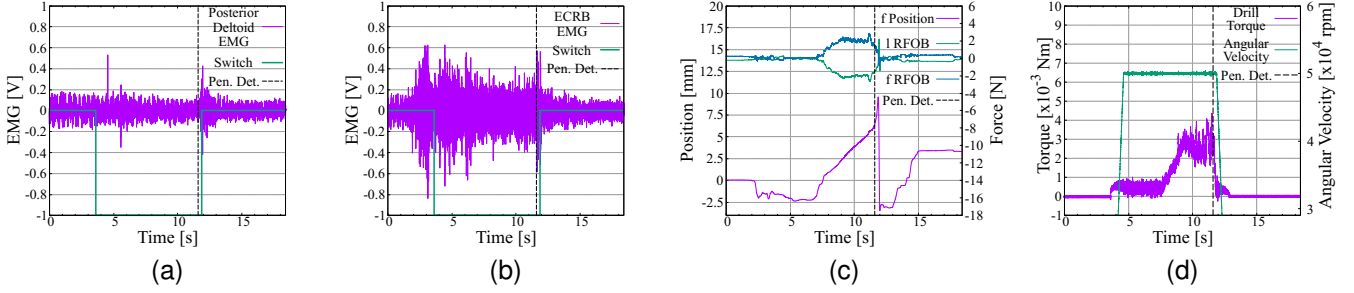


Fig. 8. Results of the drill and the electrodes in the absence of penetration detection. (a) Raw EMG values for the posterior deltoid. (b) Raw EMG values for the ECRB. (c) Position and force of the leader and follower units. (d) The torque and angular velocity of the drill bit.

spike (Fig. 8(b)), likely reflecting a rapid response to prevent unintended wrist flexion. Together, the raw EMG signals of the posterior deltoid and the ECRB indicate an increase in muscle demand when penetration detection was not operational.

B. Root Mean Square Comparison

Fig. 9 presents the normalized RMS values with a window of 2 seconds from 25 trials of operating the haptic drill with and without penetration detection, as well as using a commercial drill. These trials were all conducted by the same individual. A clear difference was observed between the three conditions; specifically, the RMS values of the EMG signals were higher when penetration detection was disabled for both the posterior deltoid and the ECRB. Furthermore, the %MVC values were clearly higher in the drilling trials conducted with the commercial drill for both muscles. This was likely because it was handheld and required more muscle stability to operate.

On the other hand, paired-sample t-tests were conducted to better examine differences in muscle activity between the two conditions using the haptic drill. For the posterior deltoid, as shown in Fig. 9(a), the RMS values were notably higher in the absence of penetration detection. The mean difference was 11.96 (SD = 6.31), and the difference was statistically significant, with $t(24) = 9.43$, $p < 0.001$. This observed change likely reflected greater motor unit recruitment as additional effort was required to pull the arm back. Users relied solely on haptic feedback to prevent puncturing through the bone.

For the ECRB, shown in Fig. 9(b), the RMS values were also higher when penetration detection condition was disabled, although the difference was not as large as that of the posterior deltoid. The mean difference was 3.96 (SD = 5.57), and

the difference was statistically significant with $t(24) = 3.56$, $p < 0.002$. Unlike the posterior deltoid, the ECRB was already engaged when holding the drill and pressing the button down. Because the muscle was contributing to wrist stabilization and grip throughout the experiment, its baseline activation was already elevated. Therefore, the release of the button increased muscle activation, but the relative change was smaller. In contrast, the posterior deltoid began with minimal activity and required increased activation to retract the shoulder, resulting in a larger observed difference between the conditions.

V. CONCLUSION

This study evaluated the impact of penetration detection in an orthopedic haptic drill from a surgeon's physiological standpoint using sEMG. By comparing muscle activation during drilling with and without penetration detection, this study demonstrated that the integration of this feature reduces muscular demand, particularly in the posterior deltoid. The muscle activation examined through raw EMG was visibly lower when the function was enabled. Furthermore, the normalized RMS values consistently showed lower EMG activity with the breach detection feature, indicating a decrease in muscle recruitment during the procedure.

As the system automatically identifies bone penetration, users do not have to rely solely on their manual judgment; thus, the application of the penetration detection feature reduces the muscular effort that users exert to control and stop the drilling process. The results of this study predict and support the possibility that using the drill with penetration detection will lessen muscle fatigue over time. Considering the prevalence of musculoskeletal disorders among surgeons today, reducing

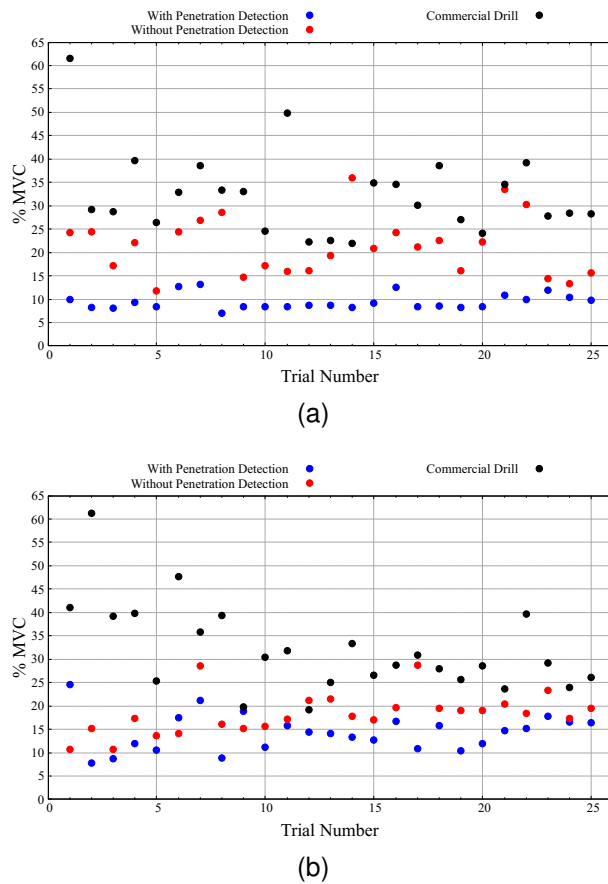


Fig. 9. The normalized RMS values of sEMG signals measured from 25 trials completed with and without penetration detection in the haptic drill, as well as with a commercially available drill. (a) The % MVC values for the posterior deltoid EMG. (b) The % MVC values for the ECRB EMG.

muscular strain may improve both their short-term comfort as well as their well-being and occupational health.

Future work should address the limitations of the current experimental validation by increasing the sample size with more participants to improve the reliability and generalizability of these findings. In addition, statistical analysis should be performed, especially between-subject analysis, to better evaluate differences and confirm the significance of reduced muscle activity. Lastly, as surgeries often take multiple hours, longer-duration experiments should be conducted to simulate such conditions. By doing such experiments, EMG signals can be analyzed by using frequency-domain analysis, providing a deeper insight into fatigue progression over time.

REFERENCES

- [1] G. Xia, Z. Jiang, and Y. Dai, "Pedicule drilling force control of a robotic surgical system via spine-soft tissue coupling model and parameters optimization," *Computers in Biology and Medicine*, vol. 169, p. 107710, Feb. 2024.
- [2] N. Vartiainen, M. Niemelä, and E.-P. Pälvimäki, "Evaluation of a novel high speed burr tip for safe and efficient bone surgery in a sheep model," *Scientific Reports*, vol. 15, no. 1, May 2025.
- [3] T. Osa, C. F. Abawi, N. Sugita, H. Chikuda, S. Sugita, T. Tanaka, H. Oshima, T. Moro, S. Tanaka and M. Mitsuishi, "Hand-Held Bone Cutting Tool With Autonomous Penetration Detection for Spinal Surgery," *IEEE/ASME Trans. Mechatronics*, vol. 20, no. 6, pp. 3018–3027, Dec. 2015.
- [4] Yamanouchi, K., Takano, S., Mima, Y., Matsunaga, T., Ohnishi, K., Matsumoto, M., Nakamura, M., Shimono, T., and Yagi, M., "Validation of a surgical drill with a haptic interface in spine surgery," *Scientific Reports*, vol. 13, no. 1, Art. 598, Jan. 2023.
- [5] M. Timmermans, A. Massalimova, R. Li, A. Davoodi, Q. Goossens, K. Niu, E. Vander Poorten, P. Färnstahl, and K. Denis, "State-of-the-Art of Non-Radiative, Non-Visual Spine Sensing with a Focus on Sensing Forces, Vibrations and Bioelectrical Properties: A Systematic Review," *Sensors*, vol. 23, no. 19, Art. 8094, Sep. 2023.
- [6] J. Merbah, B. R. Caré, P. Gorce, F. Gadea, and F. Prince, "A new approach to quantifying muscular fatigue using wearable EMG sensors during surgery: An ergonomic case study," *Sensors*, vol. 23, no. 3, Art. 1686, Feb. 2023.
- [7] G. Mavroun, T. R. Meling, J. Lafuente, K. N. Fountas, and A. K. Demetriades, "Postural ergonomics and work-related musculoskeletal disorders in neurosurgery: lessons from an international survey," *Acta Neurochirurgica*, vol. 163, no. 6, pp. 1541–1552, Feb. 2021.
- [8] V. Zakeri and A. J. Hodgson, "Automatic Identification of Hard and Soft Bone Tissues by Analyzing Drilling Sounds," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 27, no. 2, pp. 404–414, Feb. 2019.
- [9] A. Swamy, G. Burström, J. W. Spliethoff, D. Babic, C. Reich, J. Groen, E. Edström, A. E. Terander, J. M. Racadio, J. Dankelman, and B. H. W. Hendriks, "Diffuse reflectance spectroscopy, a potential optical sensing technology for the detection of cortical breaches during spinal screw placement," *J. Biomed. Opt.*, vol. 24, no. 1, pp. 1–11, Jan. 2019.
- [10] L. Li, S. Yang, W. Peng, H. Ding and G. Wang, "A CT Image-Based Virtual Sensing Method to Estimate Bone Drilling Force for Surgical Robots," *IEEE Transactions on Biomedical Engineering*, vol. 69, no. 2, pp. 871–881, Feb. 2022.
- [11] S. Takano et al., "Effectiveness Evaluation of Penetration-Detection Scheme for Surgical Drill Using Thrust Force and Positional Information," *IEEE Access*, vol. 13, pp. 125594–125605, 2025.
- [12] Y. Du, R. Jiang, and H. Wang, "Ergonomic design and assessment of an improved handle for a laparoscopic dissector based on 3D anthropometry," *International Journal of Environmental Research and Public Health*, vol. 20, no. 3, p. 2361, Jan. 29, 2023.
- [13] E. Farago, D. MacIsaac, M. Suk and A. D. C. Chan, "A Review of Techniques for Surface Electromyography Signal Quality Analysis," *IEEE Reviews in Biomedical Engineering*, vol. 16, pp. 472–486, 2023.
- [14] D. Yadav and K. Veer, "Recent trends and challenges of surface electromyography in prosthetic applications," *Biomedical Engineering Letters*, vol. 13, no. 3, pp. 353–373, Apr. 22, 2023.
- [15] H. Nakamura, "Back to Basics Series: Biomedical Measurement 1 (Electromyography)," *Transactions of Japanese Society for Medical and Biological Engineering*, vol. 57, no. 4–5, pp. 149–157, Sept. 10, 2019.
- [16] S. Muceli and R. Merletti, "Tutorial. Frequency analysis of the surface EMG signal: Best practices," *Journal of Electromyography and Kinesiology*, vol. 79, p. 102937, Dec. 2024.
- [17] P. Ortega-Auriol, T. Besier, and A. J. C. McMorland, "Effect of surface electromyography normalisation methods over gait muscle synergies," *Journal of Electromyography and Kinesiology*, vol. 80, p. 102968, Feb. 2025.
- [18] S. Fuentes del Toro and J. Aranda-Ruiz, "The Impact of Normalization Procedures on Surface Electromyography (sEMG) Data Integrity: A Study of Bicep and Tricep Muscle Signal Analysis," *Sensors*, vol. 25, no. 9, p. 2668, Apr. 23, 2025.
- [19] K. Zhao, Z. Zhang, H. Wen, and A. Scano, "Intra-Subject and Inter-Subject Movement Variability Quantified with Muscle Synergies in Upper-Limb Reaching Movements," *Biomimetics*, vol. 6, no. 4, p. 63, Oct. 20, 2021.
- [20] A. Dwivedi, D. Shieff, A. Turner, G. Gorjup, Y. Kwon and M. Liarokapis, "A Shared Control Framework for Robotic Telemanipulation Combining Electromyography Based Motion Estimation and Compliance Control," *2021 IEEE International Conference on Robotics and Automation (ICRA)*, Xi'an, China, 2021, pp. 9467–9473.
- [21] W. Guo, Z. Zhao, Z. Zhou, Y. Fang, Y. Yu, and X. Sheng, "Hand kinematics, high-density sEMG comprising forearm and far-field potentials for motion intent recognition," *Scientific Data*, vol. 12, art. no. 445, 2025.